

```
In [6]: import pandas as pd
transaction_data= pd.read_csv("C:/Users/lavanya/Downloads/QVI_transaction_data.c
customer_data=pd.read_csv("C:/Users/lavanya/Downloads/QVI_purchase_behaviour.csv

print(customer_data)
```

	LYLTY_CARD_NBR		LIFESTAGE	PREMIUM_CUSTOMER
0	1000	YOUNG	SINGLES/COUPLES	Premium
1	1002	YOUNG	SINGLES/COUPLES	Mainstream
2	1003		YOUNG FAMILIES	Budget
3	1004	OLDER	SINGLES/COUPLES	Mainstream
4	1005	MIDAGE	SINGLES/COUPLES	Mainstream
...	...		...	...
72632	2370651	MIDAGE	SINGLES/COUPLES	Mainstream
72633	2370701		YOUNG FAMILIES	Mainstream
72634	2370751		YOUNG FAMILIES	Premium
72635	2370961		OLDER FAMILIES	Budget
72636	2373711	YOUNG	SINGLES/COUPLES	Mainstream

[72637 rows x 3 columns]

```
In [ ]:
```

```
print(transaction_data)
```

```
In [9]: transaction_data['DATE']=pd.to_datetime(transaction_data['DATE'],unit='D',origin
print(transaction_data['DATE'].head())
```

```
0    2018-10-17
1    2019-05-14
2    2019-05-20
3    2018-08-17
4    2018-08-18
Name: DATE, dtype: datetime64[ns]
```

```
In [10]: unique_products=transaction_data['PROD_NAME'].unique()
for p in sorted(unique_products):
    print(p)
```

Burger Rings 220g
 CCs Nacho Cheese 175g
 CCs Original 175g
 CCs Tasty Cheese 175g
 Cheetos Chs & Bacon Balls 190g
 Cheetos Puffs 165g
 Cheezels Cheese 330g
 Cheezels Cheese Box 125g
 Cobs Popd Sea Salt Chips 110g
 Cobs Popd Sour Crm &Chives Chips 110g
 Cobs Popd Swt/Chlli &Sr/Cream Chips 110g
 Dorito Corn Chp Supreme 380g
 Doritos Cheese Supreme 330g
 Doritos Corn Chip Mexican Jalapeno 150g
 Doritos Corn Chip Southern Chicken 150g
 Doritos Corn Chips Cheese Supreme 170g
 Doritos Corn Chips Nacho Cheese 170g
 Doritos Corn Chips Original 170g
 Doritos Mexicana 170g
 Doritos Salsa Medium 300g
 Doritos Salsa Mild 300g
 French Fries Potato Chips 175g
 Grain Waves Sweet Chilli 210g
 Grain Waves Sour Cream&Chives 210G
 GrnWves Plus Btroot & Chilli Jam 180g
 Infuzions BBQ Rib Prawn Crackers 110g
 Infuzions Mango Chutny Papadums 70g
 Infuzions SourCream&Herbs Veg Strws 110g
 Infuzions Thai SweetChili PotatoMix 110g
 Infzns Crn Crnchers Tangy Gcamole 110g
 Kettle 135g Swt Pot Sea Salt
 Kettle Chilli 175g
 Kettle Honey Soy Chicken 175g
 Kettle Mozzarella Basil & Pesto 175g
 Kettle Original 175g
 Kettle Sea Salt And Vinegar 175g
 Kettle Sensations BBQ&Maple 150g
 Kettle Sensations Camembert & Fig 150g
 Kettle Sensations Siracha Lime 150g
 Kettle Sweet Chilli And Sour Cream 175g
 Kettle Tortilla ChpsBtroot&Ricotta 150g
 Kettle Tortilla ChpsFeta&Garlic 150g
 Kettle Tortilla ChpsHny&Jlpno Chili 150g
 NCC Sour Cream & Garden Chives 175g
 Natural Chip Compny SeaSalt175g
 Natural Chip Co Tmato Hrb&Spce 175g
 Natural ChipCo Hony Soy Chckn175g
 Natural ChipCo Sea Salt & Vinegr 175g
 Old El Paso Salsa Dip Chnky Tom Ht300g
 Old El Paso Salsa Dip Tomato Med 300g
 Old El Paso Salsa Dip Tomato Mild 300g
 Pringles Barbeque 134g
 Pringles Chicken Salt Crips 134g
 Pringles Mystery Flavour 134g
 Pringles Original Crisps 134g
 Pringles Slt Vingar 134g
 Pringles SourCream Onion 134g
 Pringles Sthrn FriedChicken 134g
 Pringles Sweet&Spcy BBQ 134g
 RRD Chilli& Coconut 150g

RRD Honey Soy	Chicken 165g
RRD Lime & Pepper	165g
RRD Pc Sea Salt	165g
RRD SR Slow Rst	Pork Belly 150g
RRD Salt & Vinegar	165g
RRD Steak &	Chimuchurri 150g
RRD Sweet Chilli &	Sour Cream 165g
Red Rock Deli Chikn&	Garlic Aioli 150g
Red Rock Deli SR	Salsa & Mzzrlla 150g
Red Rock Deli Sp	Salt & Truffle 150G
Red Rock Deli Thai	Chilli&Lime 150g
Smith Crinkle Cut	Bolognese 150g
Smith Crinkle Cut	Mac N Cheese 150g
Smiths Chip Thinly	Cut Original 175g
Smiths Chip Thinly	CutSalt/Vinegr175g
Smiths Chip Thinly	S/Cream&Onion 175g
Smiths Crinkle	Original 330g
Smiths Crinkle Chips	Salt & Vinegar 330g
Smiths Crinkle Cut	Chips Barbecue 170g
Smiths Crinkle Cut	Chips Chicken 170g
Smiths Crinkle Cut	Chips Chs&Onion170g
Smiths Crinkle Cut	Chips Original 170g
Smiths Crinkle Cut	French OnionDip 150g
Smiths Crinkle Cut	Salt & Vinegar 170g
Smiths Crinkle Cut	Snag&Sauce 150g
Smiths Crinkle Cut	Tomato Salsa 150g
Smiths Crnkle Chip	Orgnl Big Bag 380g
Smiths Thinly	Swt Chli&S/Cream175G
Smiths Thinly Cut	Roast Chicken 175g
Snbts Whlgrn Crisps	Cheddr&Mstrd 90g
Sunbites Whlegren	Crisps Frch/Onin 90g
Thins Chips	Originl saltd 175g
Thins Chips Light&	Tangy 175g
Thins Chips Salt &	Vinegar 175g
Thins Chips Seasoned	chicken 175g
Thins Potato Chips	Hot & Spicy 175g
Tostitos Lightly	Salted 175g
Tostitos Smoked	Chipotle 175g
Tostitos Splash Of	Lime 175g
Twisties Cheese	270g
Twisties Cheese	Burger 250g
Twisties Chicken	270g
Tyrrells Crisps	Ched & Chives 165g
Tyrrells Crisps	Lightly Salted 165g
WW Crinkle Cut	Chicken 175g
WW Crinkle Cut	Original 175g
WW D/Style Chip	Sea Salt 200g
WW Original Corn	Chips 200g
WW Original Stacked	Chips 160g
WW Sour Cream &Onion	Stacked Chips 160g
WW Supreme Cheese	Corn Chips 200g
Woolworths Cheese	Rings 190g
Woolworths Medium	Salsa 300g
Woolworths Mild	Salsa 300g

```
In [11]: transaction_data=transaction_data[~transaction_data['PROD_NAME'].str.contains('s
print(transaction_data.shape)
```

(246742, 8)

```
In [14]: print(transaction_data['PROD_QTY'].describe())
```

```
count    246742.000000
mean       1.908062
std        0.659831
min         1.000000
25%        2.000000
50%        2.000000
75%        2.000000
max        200.000000
Name: PROD_QTY, dtype: float64
```

```
In [15]: outliers=transaction_data[transaction_data['PROD_QTY']==200]
print(outliers)
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	\
69762	2018-08-19	226	226000	226201	4	
69763	2019-05-20	226	226000	226210	4	

	PROD_NAME	PROD_QTY	TOT_SALES
69762	Dorito Corn Chp Supreme 380g	200	650.0
69763	Dorito Corn Chp Supreme 380g	200	650.0

```
In [19]: transaction_data=transaction_data[transaction_data['LYLTY_CARD_NBR'] != 226000]
print(transaction_data.shape)
print(transaction_data['PROD_QTY'].describe())
```

```
(246740, 8)
count    246740.000000
mean       1.906456
std        0.342499
min         1.000000
25%        2.000000
50%        2.000000
75%        2.000000
max         5.000000
Name: PROD_QTY, dtype: float64
```

```
In [22]: transaction_data['PACK_SIZE']= transaction_data['PROD_NAME'].str.extract(r'(\d+)')
print(transaction_data['PACK_SIZE'].describe())
```

```
count    240676.000000
mean      175.302286
std       60.014468
min       70.000000
25%      150.000000
50%      170.000000
75%      175.000000
max      380.000000
Name: PACK_SIZE, dtype: float64
```

```
In [25]: transaction_data['BRAND_NAME']=transaction_data['PROD_NAME'].str.extract(r'^(\S+)')
print(sorted (transaction_data['BRAND_NAME'].unique()))
```

```
['BURGER', 'CCS', 'CHEETOS', 'CHEEZELS', 'COBS', 'DORITO', 'DORITOS', 'FRENCH',
'GRAIN', 'GRNWVES', 'INFUZIONI', 'INFZNS', 'KETTLE', 'NATURAL', 'NCC', 'PRINGLE
S', 'RED', 'RRD', 'SMITH', 'SMITHS', 'SNBTS', 'SUNBITES', 'THINS', 'TOSTITOS', 'T
WISTIES', 'TYRRELLS', 'WOOLWORTHS', 'WW']
```

```
In [26]: brand_fixes = {
        'DORITO': 'DORITOS',
```

```

    'RED': 'RRD',
    'SMITH': 'SMITHS',
    'SNBTS': 'SUNBITES',
    'INFZNS': 'INFUZIONI',
    'GRAIN': 'GRNWVES',
    'NCC': 'NATURAL',
    'WW': 'WOOLWORTHS'
}
transaction_data['BRAND_NAME'] = transaction_data['BRAND_NAME'].replace(brand_fi
print(sorted(transaction_data['BRAND_NAME'].unique()))

```

```

['BURGER', 'CCS', 'CHEETOS', 'CHEEZELS', 'COBS', 'DORITOS', 'FRENCH', 'GRNWVES',
'INFUZIONI', 'KETTLE', 'NATURAL', 'PRINGLES', 'RRD', 'SMITHS', 'SUNBITES', 'THIN
S', 'TOSTITOS', 'TWISTIES', 'TYRRELLS', 'WOOLWORTHS']

```

```

In [28]: data=transaction_data.merge(customer_data, on='LYLTY_CARD_NBR',how='left')
print(data.shape)
print(data.isnull().sum())

```

```

(246740, 12)
DATE                0
STORE_NBR           0
LYLTY_CARD_NBR      0
TXN_ID              0
PROD_NBR            0
PROD_NAME           0
PROD_QTY            0
TOT_SALES           0
PACK_SIZE           6064
BRAND_NAME           0
LIFESTAGE           0
PREMIUM_CUSTOMER    0
dtype: int64

```

```

In [32]: print(data[data['PACK_SIZE'].isnull()]['PROD_NAME'].unique())

```

```

['Grain Waves Sour      Cream&Chives 210G'
'Red Rock Deli Sp      Salt & Truffle 150G'
'Smiths Thinly          Swt Chli&S/Cream175G']

```

```

In [33]: transaction_data['PACK_SIZE'] = transaction_data['PROD_NAME'].str.extract(r'(\d+

# Re-run the merge with the corrected data
data = transaction_data.merge(customer_data, on='LYLTY_CARD_NBR', how='left')
print(data['PACK_SIZE'].isnull().sum())

```

```

0

```

```

In [34]: sales_summary = data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['TOT_SALES'].sum
sales_summary = sales_summary.sort_values('TOT_SALES', ascending=False)
print(sales_summary)

```

		LIFESTAGE	PREMIUM_CUSTOMER	TOT_SALES
6		OLDER FAMILIES	Budget	156863.75
19	YOUNG	SINGLES/COUPLES	Mainstream	147582.20
13		RETIREEES	Mainstream	145168.95
15		YOUNG FAMILIES	Budget	129717.95
9	OLDER	SINGLES/COUPLES	Budget	127833.60
10	OLDER	SINGLES/COUPLES	Mainstream	124648.50
11	OLDER	SINGLES/COUPLES	Premium	123537.55
12		RETIREEES	Budget	105916.30
7		OLDER FAMILIES	Mainstream	96413.55
14		RETIREEES	Premium	91296.65
16		YOUNG FAMILIES	Mainstream	86338.25
1	MIDAGE	SINGLES/COUPLES	Mainstream	84734.25
17		YOUNG FAMILIES	Premium	78571.70
8		OLDER FAMILIES	Premium	75242.60
18	YOUNG	SINGLES/COUPLES	Budget	57122.10
2	MIDAGE	SINGLES/COUPLES	Premium	54443.85
20	YOUNG	SINGLES/COUPLES	Premium	39052.30
0	MIDAGE	SINGLES/COUPLES	Budget	33345.70
3		NEW FAMILIES	Budget	20607.45
4		NEW FAMILIES	Mainstream	15979.70
5		NEW FAMILIES	Premium	10760.80

```
In [37]: sales_summary = data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['TOT_SALES'].sum
sales_summary = sales_summary.sort_values('TOT_SALES', ascending=False)
print(sales_summary)
customer_summary = data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['LYLTY_CARD_N
print(customer_summary.sort_values('CUSTOMERS', ascending=False))
avg_spend = data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER']).apply(
    lambda x: x['TOT_SALES'].sum() / x['LYLTY_CARD_NBR'].nunique()
).reset_index(name='AVG_SPEND_PER_CUSTOMER')
print(avg_spend.sort_values('AVG_SPEND_PER_CUSTOMER', ascending=False))
```

		LIFESTAGE	PREMIUM_CUSTOMER	TOT_SALES
6		OLDER FAMILIES	Budget	156863.75
19	YOUNG	SINGLES/COUPLES	Mainstream	147582.20
13		RETIREEES	Mainstream	145168.95
15		YOUNG FAMILIES	Budget	129717.95
9	OLDER	SINGLES/COUPLES	Budget	127833.60
10	OLDER	SINGLES/COUPLES	Mainstream	124648.50
11	OLDER	SINGLES/COUPLES	Premium	123537.55
12		RETIREEES	Budget	105916.30
7		OLDER FAMILIES	Mainstream	96413.55
14		RETIREEES	Premium	91296.65
16		YOUNG FAMILIES	Mainstream	86338.25
1	MIDAGE	SINGLES/COUPLES	Mainstream	84734.25
17		YOUNG FAMILIES	Premium	78571.70
8		OLDER FAMILIES	Premium	75242.60
18	YOUNG	SINGLES/COUPLES	Budget	57122.10
2	MIDAGE	SINGLES/COUPLES	Premium	54443.85
20	YOUNG	SINGLES/COUPLES	Premium	39052.30
0	MIDAGE	SINGLES/COUPLES	Budget	33345.70
3		NEW FAMILIES	Budget	20607.45
4		NEW FAMILIES	Mainstream	15979.70
5		NEW FAMILIES	Premium	10760.80
		LIFESTAGE	PREMIUM_CUSTOMER	CUSTOMERS
19	YOUNG	SINGLES/COUPLES	Mainstream	7917
13		RETIREEES	Mainstream	6358
10	OLDER	SINGLES/COUPLES	Mainstream	4858
9	OLDER	SINGLES/COUPLES	Budget	4849
11	OLDER	SINGLES/COUPLES	Premium	4682
6		OLDER FAMILIES	Budget	4611
12		RETIREEES	Budget	4385
15		YOUNG FAMILIES	Budget	3953
14		RETIREEES	Premium	3812
18	YOUNG	SINGLES/COUPLES	Budget	3647
1	MIDAGE	SINGLES/COUPLES	Mainstream	3298
7		OLDER FAMILIES	Mainstream	2788
16		YOUNG FAMILIES	Mainstream	2685
20	YOUNG	SINGLES/COUPLES	Premium	2480
17		YOUNG FAMILIES	Premium	2398
2	MIDAGE	SINGLES/COUPLES	Premium	2369
8		OLDER FAMILIES	Premium	2231
0	MIDAGE	SINGLES/COUPLES	Budget	1474
3		NEW FAMILIES	Budget	1087
4		NEW FAMILIES	Mainstream	830
5		NEW FAMILIES	Premium	575
		LIFESTAGE	PREMIUM_CUSTOMER	AVG_SPEND_PER_CUSTOMER
7		OLDER FAMILIES	Mainstream	34.581618
6		OLDER FAMILIES	Budget	34.019464
8		OLDER FAMILIES	Premium	33.725952
15		YOUNG FAMILIES	Budget	32.815065
17		YOUNG FAMILIES	Premium	32.765513
16		YOUNG FAMILIES	Mainstream	32.155773
11	OLDER	SINGLES/COUPLES	Premium	26.385636
9	OLDER	SINGLES/COUPLES	Budget	26.362879
1	MIDAGE	SINGLES/COUPLES	Mainstream	25.692617
10	OLDER	SINGLES/COUPLES	Mainstream	25.658399
12		RETIREEES	Budget	24.154230
14		RETIREEES	Premium	23.949803
2	MIDAGE	SINGLES/COUPLES	Premium	22.981786
13		RETIREEES	Mainstream	22.832487
0	MIDAGE	SINGLES/COUPLES	Budget	22.622592

4	NEW FAMILIES	Mainstream	19.252651
3	NEW FAMILIES	Budget	18.958096
5	NEW FAMILIES	Premium	18.714435
19	YOUNG SINGLES/COUPLES	Mainstream	18.641177
20	YOUNG SINGLES/COUPLES	Premium	15.746895
18	YOUNG SINGLES/COUPLES	Budget	15.662764

C:\Users\lavanya\AppData\Local\Temp\ipykernel_1316\1909808970.py:6: FutureWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

```
avg_spend = data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER']).apply(
```

```
In [38]: avg_pack=data.groupby(['LIFESTAGE', 'PREMIUM_CUSTOMER'])['PACK_SIZE'].mean().reset_index()
print(avg_pack.sort_values('PACK_SIZE',ascending=False))
```

	LIFESTAGE	PREMIUM_CUSTOMER	PACK_SIZE
19	YOUNG SINGLES/COUPLES	Mainstream	178.344249
1	MIDAGE SINGLES/COUPLES	Mainstream	177.898693
11	OLDER SINGLES/COUPLES	Premium	176.485568
12	RETIREES	Budget	176.395641
14	RETIREES	Premium	176.368421
4	NEW FAMILIES	Mainstream	175.629748
6	OLDER FAMILIES	Budget	175.546342
15	YOUNG FAMILIES	Budget	175.459720
9	OLDER SINGLES/COUPLES	Budget	175.334673
5	NEW FAMILIES	Premium	175.245296
13	RETIREES	Mainstream	175.213671
7	OLDER FAMILIES	Mainstream	175.175666
10	OLDER SINGLES/COUPLES	Mainstream	174.812145
3	NEW FAMILIES	Budget	174.766643
17	YOUNG FAMILIES	Premium	174.659032
2	MIDAGE SINGLES/COUPLES	Premium	174.585391
8	OLDER FAMILIES	Premium	174.524752
16	YOUNG FAMILIES	Mainstream	174.202059
0	MIDAGE SINGLES/COUPLES	Budget	174.182690
20	YOUNG SINGLES/COUPLES	Premium	174.002563
18	YOUNG SINGLES/COUPLES	Budget	173.378981

In []:

In []: